

HARRIET ALEXANDER FIELD, CO, USA

WMO: 720532

Lat: 38.533N

Lon: 106.050W

Elev: 7527

StdP: 11.11

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 00159

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 2.7	(c) 8.2	(d) -11.1	(e) 4.0	(f) 25.0	(g) -7.8	(h) 4.8	(i) 26.6	(j) 29.9	(k) 46.0	(l) 27.2	(m) 40.8	(n) 5.1	(o) 130	(p) 0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.9	89.0	53.8	86.9	53.5	84.4	53.5	59.4	73.3	58.4	72.9	57.4	72.6	10.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.0	88.6	62.3	54.4	83.6	62.1	52.5	77.9	62.4	29.7	72.7	28.9	72.5	28.2	72.5	63.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.1	(b) 23.4	(c) 20.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 26.1	(b) 23.4	(c) 20.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	31.7	32.6	42.0	46.5	54.0	67.0	70.4	67.8	62.5	50.2	39.4	31.3		
	DBStd	15.84	8.40	9.98	7.83	7.60	7.02	5.01	3.25	3.44	5.30	8.14	8.72	9.77		
	HDD50	2506	569	488	263	149	41	0	0	0	1	93	323	579		
	HDD65	5976	1034	906	712	555	341	37	3	15	103	459	767	1044		
	CDD50	2403	0	2	15	44	167	509	632	553	375	99	7	0		
	CDD65	399	0	0	0	0	2	96	170	103	27	1	0	0		
	CDH74	5590	0	0	0	7	130	1693	1892	1263	580	25	0	0		
	CDH80	1790	0	0	0	0	7	613	702	357	111	0	0	0		
	WSAvg	8.6	9.3	10.5	9.7	9.3	8.2	8.4	6.9	6.7	7.2	8.2	9.2	9.4		
	PrecAvg	11.5	0.6	0.6	0.9	1.2	1.1	0.7	1.5	1.9	1.1	0.9	0.6	0.6		
(15) Precipitation	PrecMax	14.5	0.9	1.5	1.9	3.7	3.0	1.5	3.1	4.1	2.0	2.0	1.5	2.1		
	PrecMin	6.9	0.2	0.2	0.2	0.2	0.1	0.2	0.3	0.6	0.4	0.2	0.2	0.2		
	PrecStd	2.2	0.2	0.4	0.5	0.9	0.8	0.4	0.8	0.8	0.5	0.5	0.4	0.4		
	PrecStd	2.2	0.2	0.4	0.5	0.9	0.8	0.4	0.8	0.8	0.5	0.5	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	59.4	70.3	74.6	81.1	92.6	91.2	88.9	86.8	77.3	65.9	57.2		
		MCWB	37.7	40.1	43.5	45.7	47.9	53.7	54.7	53.1	53.6	49.1	42.6	37.3		
	2%	DB	52.1	55.2	66.4	70.3	77.5	89.3	89.1	86.6	83.2	73.1	62.0	52.8		
		MCWB	35.3	37.4	42.0	44.0	46.9	53.0	54.6	53.6	51.9	46.6	41.0	35.9		
	5%	DB	48.2	51.2	62.4	66.6	74.8	86.7	86.9	84.1	80.4	70.0	58.6	49.5		
		MCWB	34.1	35.6	40.0	42.8	46.4	52.2	54.7	54.1	51.2	45.5	39.4	34.8		
	10%	DB	45.2	47.6	57.3	63.6	71.4	83.6	84.3	81.5	77.3	66.2	54.4	45.9		
		MCWB	32.4	34.5	37.7	41.5	45.7	51.2	55.0	54.4	50.6	44.7	37.4	33.8		
	0.4%	WB	39.4	40.4	44.3	47.6	52.6	59.2	61.2	60.3	58.4	52.5	44.4	39.5		
		MCDB	52.2	56.9	66.8	66.2	66.2	77.2	73.6	73.0	70.0	67.2	60.0	51.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.1	38.5	42.5	45.7	50.0	57.0	59.9	58.9	56.5	50.2	42.1	37.5		
		MCDB	48.3	52.2	63.0	66.3	67.0	74.7	73.3	72.6	68.5	66.2	58.1	48.5		
	5%	WB	35.1	36.9	40.7	44.2	48.4	55.4	59.0	57.8	55.1	47.9	40.3	36.2		
		MCDB	46.1	49.2	59.4	63.4	67.6	75.4	73.1	72.5	69.3	63.7	56.4	47.6		
	10%	WB	33.2	34.9	38.9	42.5	47.1	54.1	58.1	56.9	53.8	46.1	38.3	34.2		
		MCDB	42.9	46.7	54.9	60.4	67.0	75.2	73.0	72.1	69.4	62.7	53.2	45.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.2	21.1	25.4	26.6	27.0	31.5	28.9	28.7	28.2	26.7	23.4	20.9		
		MCDBR	25.9	23.4	31.0	31.6	32.5	34.2	32.7	32.1	32.9	31.1	27.9	24.1		
		MCWBR	14.5	12.0	13.6	13.1	12.0	11.3	9.2	9.5	11.0	12.7	13.2	12.8		
		MCDBR	22.7	21.4	28.5	29.8	29.8	30.3	27.4	26.7	25.6	26.9	26.3	20.5		
	5% WB	MCWBR	13.2	11.4	13.1	12.8	11.3	10.3	8.4	8.6	8.7	11.2	12.8	11.5		
		MCWBR	13.2	11.4	13.1	12.8	11.3	10.3	8.4	8.6	8.7	11.2	12.8	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.228	0.236	0.265	0.286	0.301	0.307	0.348	0.333	0.288	0.266	0.241	0.232		
	taud		2.524	2.502	2.484	2.491	2.517	2.514	2.451	2.466	2.603	2.621	2.605	2.537		
	Ebn at Noon		312	322	319	315	310	306	293	296	307	306	305	303		
	Edh at Noon		23	28	31	33	33	33	35	34	28	24	22	22		
(44) All-Sky Solar Radiation	RadAvg		890	1199	1610	1899	2069	2311	2030	1853	1686	1307	963	773		
	RadStd		43	68	120	117	157	136	107	87	98	86	56	45		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131	+36

Nomenclature: See separate page